LAB 6: Attention-Mechanism and Transformers

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O U T L I N E

Part 1: What is the Transformer?

Part 2: Transformers in PyTorch

- Language Translation

Part 3: Lab Assignment

- Translation

WHAT IS THE TRANSFORMER ARCHITECTURE?

Attention Recap

Sequence-to-Sequence Transformers

Training Transformers

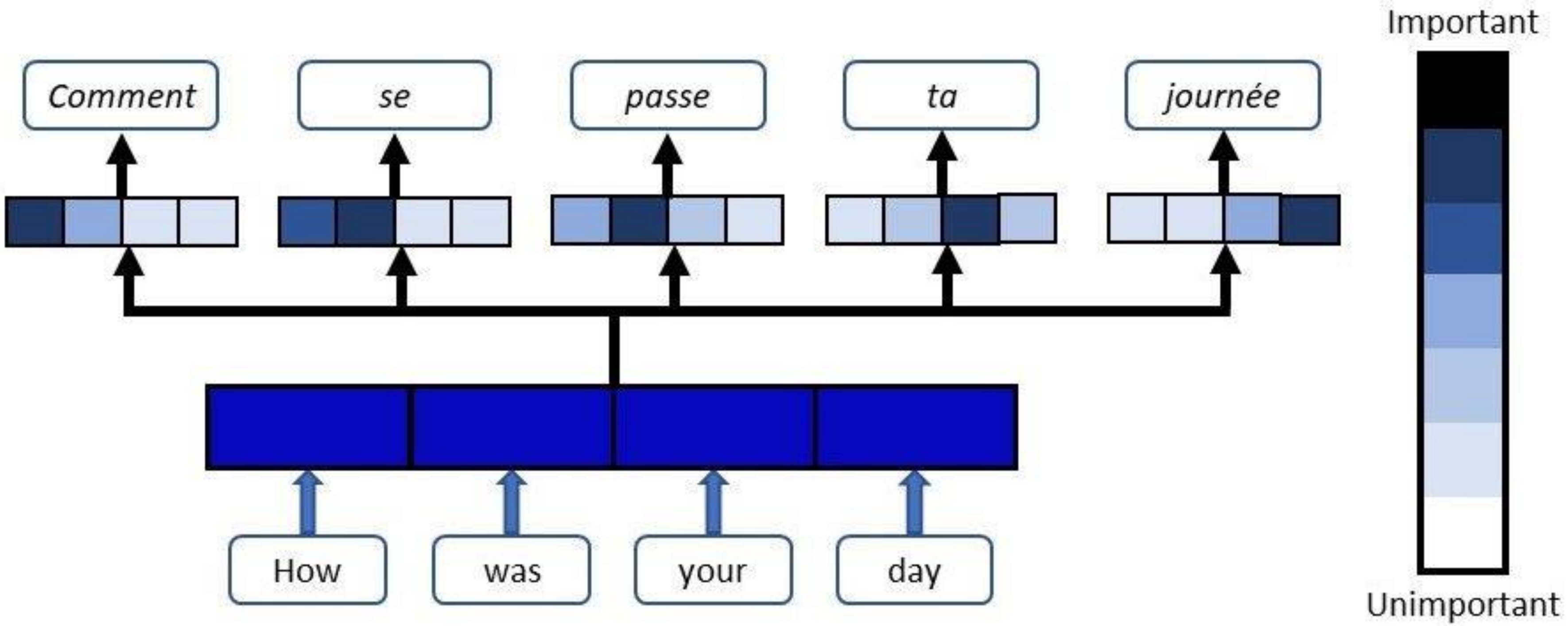
What is Attention?

What is Attention?

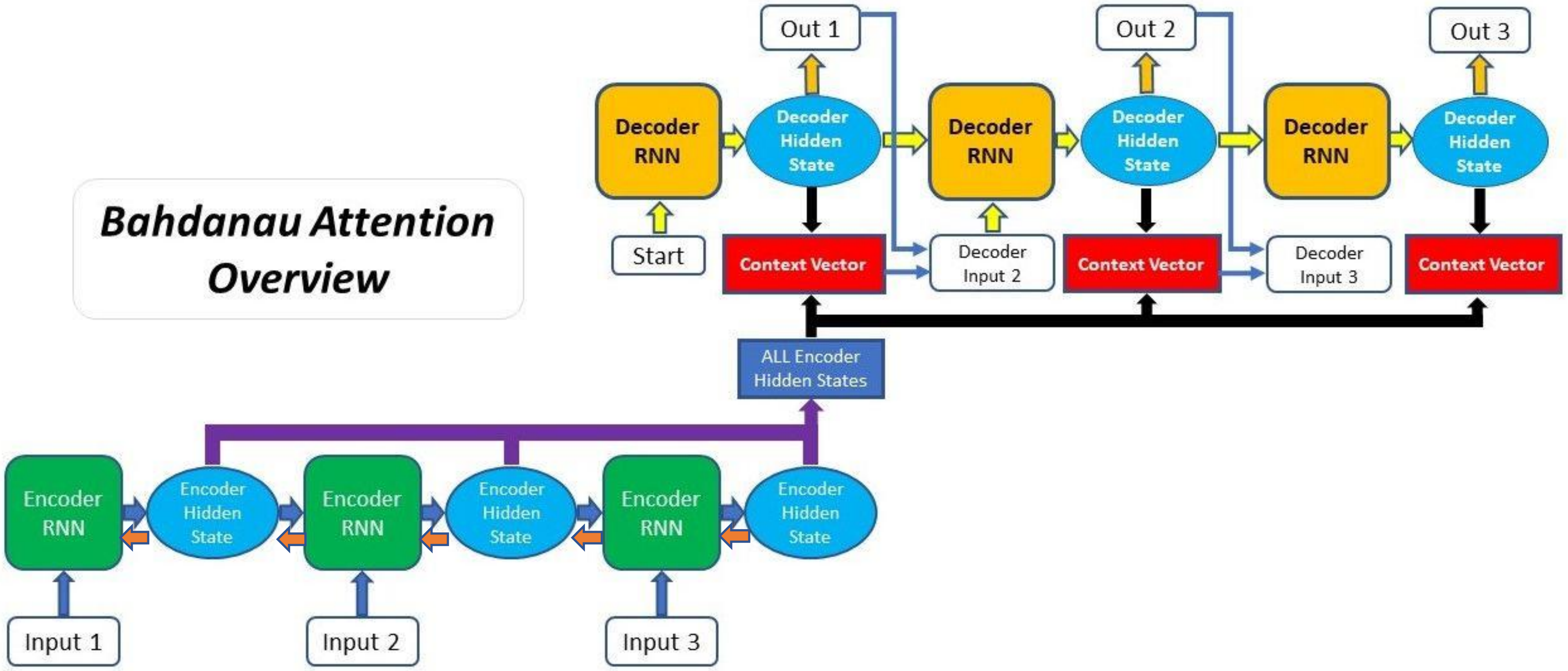
Attention in Neural Networks:

Network component (glasses) which **quantifies** the **emphasis on certain input element** when producing **output**

Attention Model

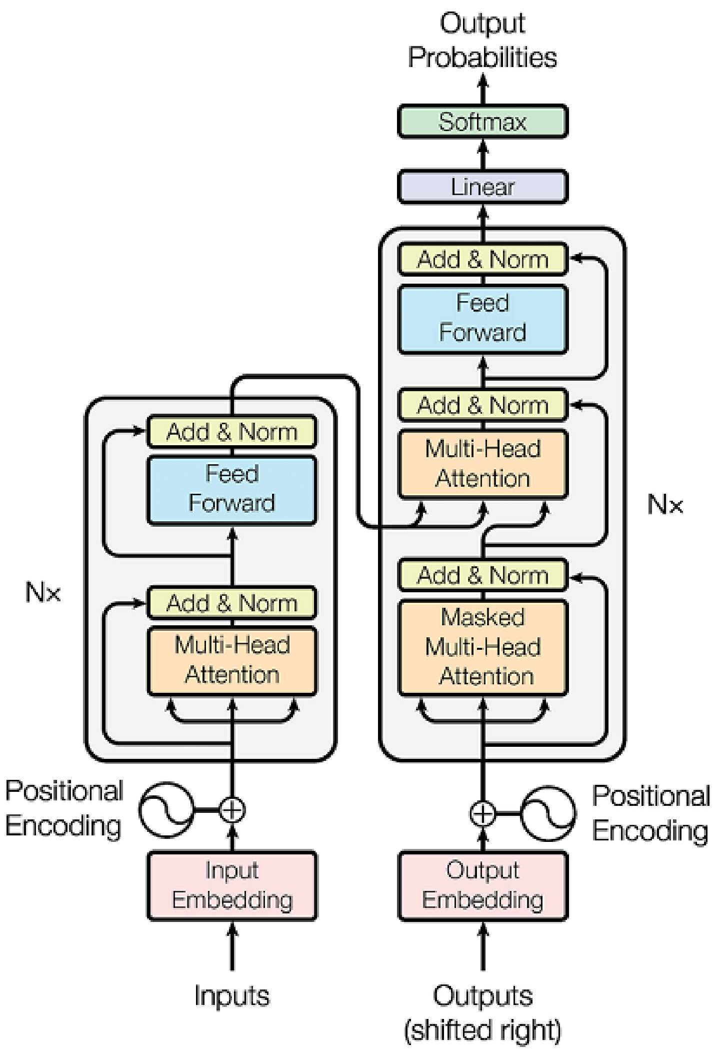


Attention Model



D. Bahdanau, K. Cho, Y. Bengio, arxiv:1409.0473

Transformer

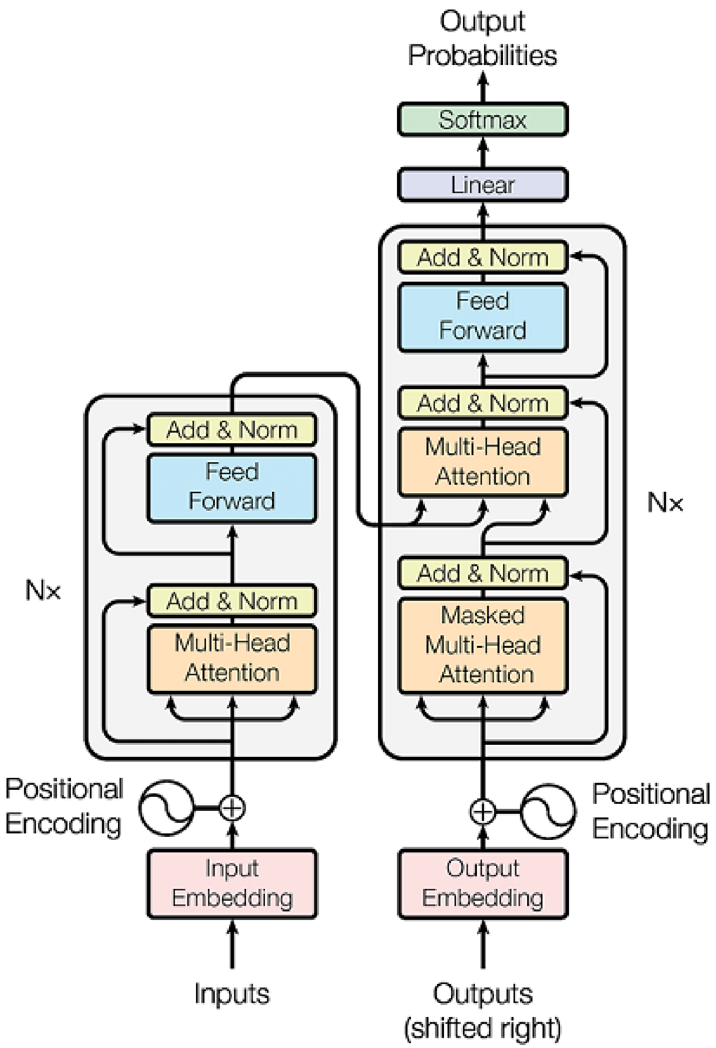


Exclusively uses attention mechanism to process sequential data

Does not necessarily process data in order like RNNs

Currently state-of-the-art method for NLP

Transformer



Attention Is All You Need

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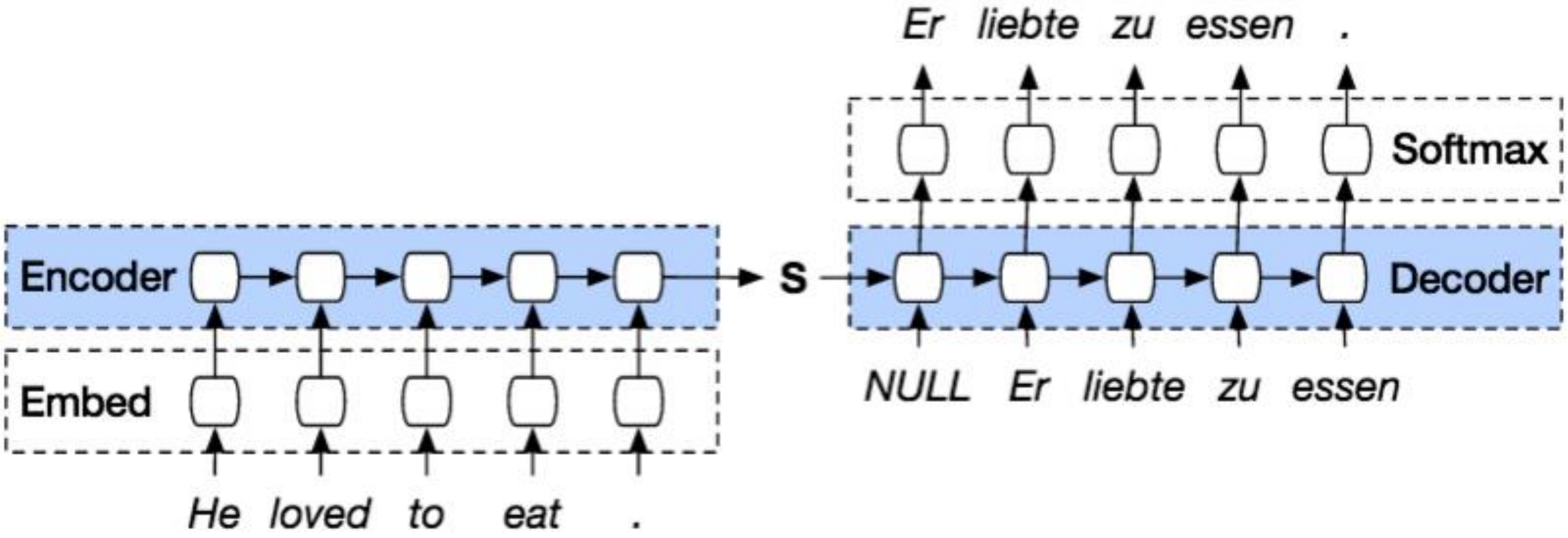
Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

2017 Neurips Proceeding

Sequence-to-Sequence Transformer



Masking

```python
# Helper
def generate_square_subsequent_mask(sz):
    mask = (torch.triu(torch.ones((sz, sz), device=DEVICE)) == 1).transpose(0, 1)
    mask = mask.float().masked_fill(mask == 0, float('-inf')).masked_fill(mask == 1, float(0.0))
    return mask

# Mask function
def create_mask(src, tgt):
    src_seq_len = src.shape[0]
    tgt_seq_len = tgt.shape[0]

    tgt_mask = generate_square_subsequent_mask(tgt_seq_len)
    src_mask = torch.zeros((src_seq_len, src_seq_len),device=DEVICE).type(torch.bool)

    src_padding_mask = (src == PAD_IDX).transpose(0, 1)
    tgt_padding_mask = (tgt == PAD_IDX).transpose(0, 1)
    return src_mask, tgt_mask, src_padding_mask, tgt_padding_mask
```

- Hides the *future* of the sequence from the current attention-head
- Hides part of the *past* of the sequence from the current attention-head

See example notebook for detailed code implementation

Collation

```python
# A helper function to club together sequential operations
def sequential_transforms(*transforms):
    def func(txt_input):
        for transform in transforms:
            txt_input = transform(txt_input)
        return txt_input
    return func

# Adds BOS/EOS and creates a tensor for input sequence indices
def tensor_transform(token_ids):
    return torch.cat((torch.tensor([BOS_IDX]),
                      torch.tensor(token_ids),
                      torch.tensor([EOS_IDX])))

# ``src`` and ``tgt`` language text transforms convert the raw strings into tensors indices
text_transform = {}
for ln in [SRC_LANGUAGE, TGT_LANGUAGE]:
    text_transform[ln] = sequential_transforms(token_transform[ln], #Tokenization
                                               vocab_transform[ln], #Numericalization
                                               tensor_transform) # Add BOS/EOS and create tensor


# The "collation" function collates data samples into batch tensors
def collate_fn(batch):
    src_batch, tgt_batch = [], []
    for src_sample, tgt_sample in batch:
        src_batch.append(text_transform[SRC_LANGUAGE](src_sample.rstrip("\n")))
        tgt_batch.append(text_transform[TGT_LANGUAGE](tgt_sample.rstrip("\n")))

    src_batch = pad_sequence(src_batch, padding_value=PAD_IDX)
    tgt_batch = pad_sequence(tgt_batch, padding_value=PAD_IDX)
    return src_batch, tgt_batch
```

- Tokenization + numericalization + batchification all in one step

See example notebook for detailed code implementation

LAB 7 ASSIGNMENT:

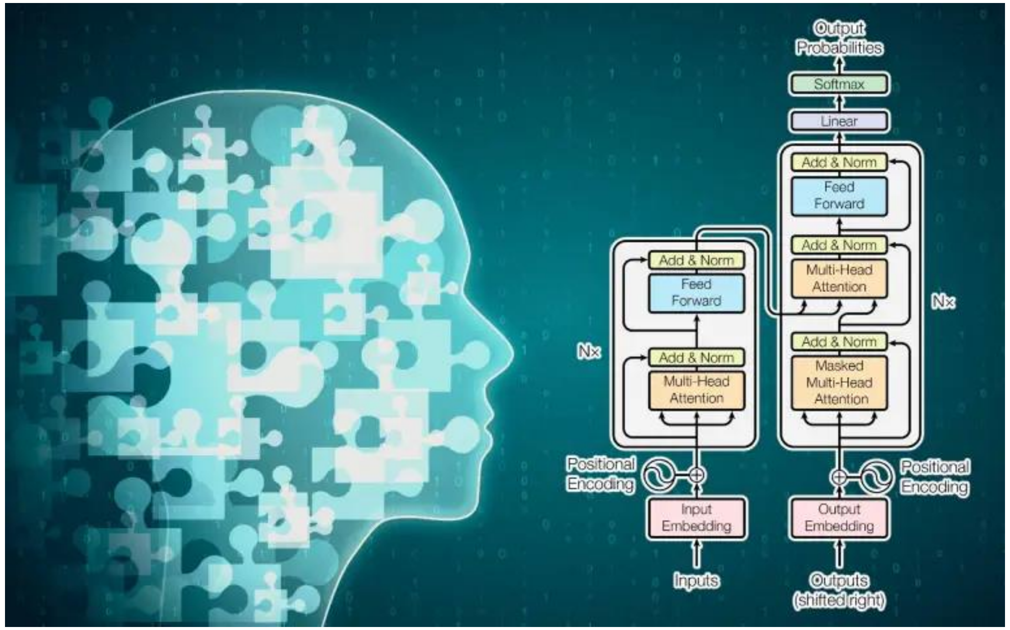
German-to-English Translation with Attention-Mechanism Transformer Model

German-to-English Dataset

Language data: image descriptions

```
Ein Mann, der mit einer Tasse Kaffee an einem Urinal steht. ==> A man standing at a urinal with a coffee cup.
Fünf gehende Personen mit einem mehrfarbigen Himmel im Hintergrund. ==> Five people walking with a multicolored sky in the background.
Ein alter Mann, der allein ein Bier trinkt. ==> A old man having a beer alone.
Ein geschulter Polizeihund sitzt neben dem Hundeführer vor dem Polizeitransporter. ==> A trained police dog sits next to his handler in front of the police van.
Ein Mann mit einem nach hinten gerichteten Hut arbeitet an Maschinen. ==> A man with a backwards hat works on machinery.
Ein asiatischer Mann kehrt den Gehweg. ==> Asian man sweeping the walkway.
Ein Mann lehnt sich in ein Auto, um mit dem Fahrer zu reden, während ein Mann auf einem Fahrrad zusieht. ==> A man leans into a car to talk to the driver, as a man on a bicycle looks on.
Zwei Kleinkinder im Freien auf dem Gras. ==> Two young toddlers outside on the grass.
Leute sehen einer Person in einem seltsamen Fahrzeug auf einem Platz zu. ==> People are watching a person in a weird vehicle in a plaza.
Ein Mann geht an einem silbernen Fahrzeug vorbei. ==> A man walks by a silver vehicle.
Eine schöne Braut geht auf einem Gehweg mit ihrem neuen Ehemann. ==> A beautiful bride walking on a sidewalk with her new husband.
Ein kleiner Junge spielt bei McDonald's GameCube. ==> A little boy playing GameCube at a McDonald's.
Ein weißer Hund schüttelt sich am Rande eines Strands mit einem orangefarbenen Ball. ==> A white dog shakes on the edge of a beach with an orange ball.
Eine Gruppe von Personen, die im Park grillen. ==> A group of people having a barbecue at a park.
Ein Mann mit Sonnenbrille legt seinen Arm um eine Frau in einer schwarz-weißen Bluse. ==> A man in sunglasses puts his arm around a woman in a black and white blouse.
Ein Mann mit einem Luftballonhut und Leute, die im Freien an Picknicktischen essen. ==> A man with a balloon hat and people eating outdoors at picnic tables.
```

Language Translation with Attention-Mechanism Transformer



In this exercise, you will use a Sequence-to-Sequence Transformer model to translate German sentences into English. Your goal is to receive an average validation (Cross-Entropy) loss of less than 2.4.

Before training, it is important to properly tokenize the data. We have worked with language data in the past, but this time the data is more complex and higher-dimensional, and so is the tokenization process. See the example lab for an in-depth look at the tokenization.

During training, we employ the same sliding-window method used in Labs 5 and 6.

After training, demonstrate your model's translation ability by comparing a few of its outputs with the ground-truth labels, along with the inputs.